

```

{
    int i;
    u8 byte;

    if (*off >= VRAM_SIZE)
    {
        return 0;
    }
    if (*off + len > VRAM_SIZE)
    {
        len = VRAM_SIZE - *off;
    }
    for (i = 0; i < len; i++)
    {
        if (copy_from_user(&byte, buf + i, 1))
        {
            return -EFAULT;
        }
        iowrite8(byte, (u8 *)vram + *off + i);
        *off += len;
    }

    return len;
}

static struct file_operations vram_fops =
{
    .owner = THIS_MODULE,
    .open = my_open,
    .release = my_close,
    .read = my_read,
    .write = my_write
};

static int __init vram_init(void) /* Constructor */
{
    if ((vram = ioremap(VRAM_BASE, VRAM_SIZE)) == NULL)
    {
        printk(KERN_ERR "Mapping video RAM failed\n");
        return -1;
    }
    if (alloc_chrdev_region(&first, 0, 1, "vram") < 0)
    {
        return -1;
    }
    if ((cl = class_create(THIS_MODULE, "chardrv")) == NULL)
    {
        unregister_chrdev_region(first, 1);
        return -1;
    }
    if (device_create(cl, NULL, first, NULL, "vram") == NULL)
    {
        class_destroy(cl);
        unregister_chrdev_region(first, 1);
    }
}

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        return -1;
    }

    cdev_init(&c_dev, &vram_fops);
    if (cdev_add(&c_dev, first, 1) == -1)
    {
        device_destroy(cl, first);
        class_destroy(cl);
        unregister_chrdev_region(first, 1);
        return -1;
    }
    return 0;
}

static void __exit vram_exit(void) /* Destructor */
{
    cdev_del(&c_dev);
    device_destroy(cl, first);
    class_destroy(cl);
    unregister_chrdev_region(first, 1);
    iounmap(vram);
}

module_init(vram_init);
module_exit(vram_exit);

MODULE_LICENSE("GPL");
MODULE_AUTHOR("Anil Kumar Pugalia <email@sarika-pugs.com>");
MODULE_DESCRIPTION("Video RAM Driver");

```

Summing up

Shweta then repeated the usual steps:

- Build the 'vram' driver (*video_ram.ko* file) by running *make* with a changed Makefile.
- Load the driver using *insmod video_ram.ko*.
- Write into */dev/vram*, say, using *echo -n "0123456789" > /dev/vram*
- Read the */dev/vram* contents using *od -t x1 -v /dev/vram | less*. (The usual *cat /dev/vram* can also be used, but that would give all the binary content. *od -t x1* shows it as hexadecimal. For more details, run *man od*.)
- Unload the driver using *rmmod video_ram*.

With half an hour still left for the end of the practical class, Shweta decided to walk around and possibly help somebody else with their experiments. **END** 

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The author is a freelance trainer in Linux internals, Linux device drivers, embedded Linux and related topics. Prior to this, he was at Intel and Nvidia. He has been working with Linux since 1994. A gold medalist from the Indian Institute of Science, Linux and knowledge sharing are two of his many passions. More about him can be found at <http://profession.sarika-pugs.com>, including information about his various Linux-related training sessions and workshops. He can be reached at email@sarika-pugs.com.